

ALWASI MUHRIJAFIATNA

Address: Bandung, Jawa Barat
Phone: +62 895-3597-76598
Email: alwasiwasi29@gmail.com
Website: alwasim.github.io

SUMMARY

Detail-oriented Electrical Engineering graduate and Certified System Analyst with hands-on experience in IoT integration, system architecture design, and Python development. Proven track record as a Maintenance & Engineering Supervisor in the manufacturing sector, adept at optimizing equipment performance and reducing machinery downtime. Possesses strong analytical skills and a commitment to driving technical solutions in cross-functional team environments.

WORK EXPERIENCE

Supervisor Maintenance & Engineering, KIA Ceramics **Mar 2025 - Sep 2025**

- Analyzed extensive maintenance history logs to identify equipment failure patterns and optimize maintenance schedules.
- Engineered data collection workflows for equipment performance metrics, generating actionable insights to strategically reduce machine downtime.
- Utilized operational data monitoring to oversee critical manufacturing machinery, driving data-backed decisions that ensured a 90% operational uptime.

Aviation Electronics Teacher, SMKN 12 Bandung **Aug 2023 - Dec 2023**

- Evaluated performance metrics of 64 students to tailor mentoring strategies in aviation navigation and communication systems, achieving a 95% class pass rate.
- Simplified complex electronics concepts.
- Developed 12 comprehensive teaching modules to simplify complex avionics concepts, boosting student practical test scores by 60%.

IoT Project Leader, PT. Inovindo digital media **Jun 2023 - Aug 2023**

- Directed a team to design and build a 'Smart Pot' IoT system aimed at optimizing the growth and care of ornamental plants.
- Drafted technical documentation and system requirements that were adopted as standard learning modules for intern students at Inovindo Academy.
- The Project is successfully integrated and utilized as core educational material for Inovindo Academy intern students.

EDUCATION

Bachelor of Education **Aug 2020 - Oct 2024**

Indonesia University of Education

- Major in Electrical Engineering Education.
- Thesis on "Application of Machine Learning and Deep Learning for the Identification of Horticultural Plants".

CERTIFICATIONS

Certified System Analyst - BNSP (Disnaker Bandung) **Aug 2026 - Aug 2029**

Test for BNSP Certificate & 200 Hour Intensive Training Program.

Certified Python AI Developer - MIMO **Lifetime Certificate**

Completed professional certification covering core Python concepts, functional and object-oriented programming, API integrations, and hands-on experience using best-in-class LLMs.

PROJECTS HISTORY

Application of Machine Learning and Deep Learning for the Identification of Horticultural Plants

- Developed a web-based plant recommendation application utilizing Python Django Framework.
- Integrated ML/DL models to process geospatial (latitude/longitude) and environmental data, delivering accurate horticultural recommendations.

IoT Smartpot System

- Engineered an ESP32-based IoT prototype integrated with Arduino Cloud, establishing a reliable data telemetry pipeline for real-time mobile monitoring.
- Led hardware-software integration to ensure seamless data flow from sensors to cloud infrastructure, successfully utilizing the system as core training material at Inovindo Academy.

Employee Complaint Application System (SRS)

- Authored comprehensive System Requirement Specification (SRS) documentation to define software architecture, user flows, and database requirements for an internal employee service platform.

CORE SKILLS

Data & System Analysis
Python, SQL/MySQL, System Requirement Specification (SRS), SDLC, Business Logic, Data Modeling/Flowcharting, Jupyter Notebook, R Studio, Matlab, Application Testing, ERP Systems.

IoT & Hardware
C++, Arduino IDE, Proteus, AutoCAD, Microcontrollers & Sensors, PLC, OPC UA Protocol, Electrical Installation, Arduino Cloud, ThingSpeak, Blynk.

Soft Skills
Leadership, Technical Documentation, Problem Solving, Cross-functional Communication, Team Collaboration, Prioritization, Time management under tight deadlines.

ORGANIZATIONAL EXPERIENCE

Electrical Engineering Student Association **Jul 2022 - Jul 2023**
Head of Telecommunication Engineering Development Sub-Division

- Designed and executed targeted work programs to enhance students' practical knowledge in the telecommunications field.

Electrical Engineering Student Association **Sept 2021 - Jan 2022**
Project Manager Workshop Telecommunication

- Managing & Lead a team to organize and execute a technical workshop

ADDITIONAL INFORMATION

Language
Indonesia (Native) & English (Advanced in Reading/Comprehension, Conversational Speaking).

Entrepreneurial & Business Experience, Kin La Fleur Eternelle (Flowershop)
Managed end-to-end business operations, digital marketing, and customer relations, demonstrating strong entrepreneurial acumen and practical business logic.

Willingness to Relocate / Availability
Willing to relocate across Indonesia & available for immediate employment.